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DISTANCE IN KN (OTHER DISTANCE METRICS)



Euclidean Distance– It is the most common distance metric and is used when data is dense or continuous. It's the default metric for many algorithms.

► Formula: $d(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \dots + (p_n - q_n)^2}$

► Python Snippet:

```
from scipy.spatial import distance  
dst = distance.euclidean(p, q)
```





```
knn = KNeighborsClassifier(n_neighbors=3, metric='euclidean')
```

Scenarios when you might want to avoid using it:

- ▶ **High Dimensionality:** Euclidean distance can be less discriminative in high-dimensional spaces.
- ▶ **Different Scales:** Varied feature scales impact distance calculations. Normalize data first.
- ▶ **Non-Linear Relationships:** Euclidean distance assumes linearity. For complex relationships, use other metrics.
- ▶ **Categorical Features:** Use Hamming distance for categorical data.
- ▶ **Outliers:** Outliers greatly affect Euclidean distance.
- ▶ **Sparse Data:** For mostly zeros, consider cosine similarity.

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Manhattan Distance (L1 norm)– Useful in high dimensional spaces, especially when dimensions are not similar in type (e.g., height vs. age). It is less influenced by outlier features.

► Formula: $d(p, q) = |p_1 - q_1| + |p_2 - q_2| + \dots + |p_n - q_n|$

► Python Snippet:

```
dst = distance.cityblock(p, q)
#or
knn = KNeighborsClassifier(n_neighbors=3, metric='manhattan')
```